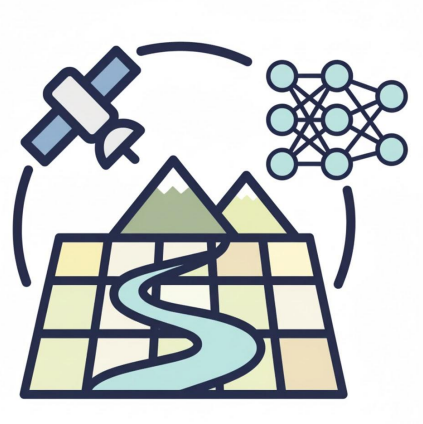




Multimodal Post-Flood Water Extent Mapping With SAR and Incomplete Multispectral Data Using a Spatially Masked Adaptive Gated Network

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I. Motivation

- Multimodal Satellite Data for Flood Mapping**
 - Accurate flood mapping is essential for effective flood management.
 - SAR** (Synthetic Aperture Radar): Enables all-weather, day-and-night observation.
 - MSI** (Multispectral Imaging): Provides high accuracy with water-sensitive bands (e.g., NIR) but is limited by weather conditions like clouds.
 - Multimodal Fusion**: Combining SAR and MSI enhances detection performance by leveraging complementary information.
- Why Deep Learning?**
 - In contrast to rule-based methods, deep learning can automatically extract and fuse spatial and spectral features.
- Challenge: Missing Data in Coregistration**
 - Coregistration, the process of aligning the coordinates of satellite data from different sensors (SAR and MSI), leads to missing values.
 - Existing DL models require data pairs without missing values and exhibit performance degradation when missing data is present.

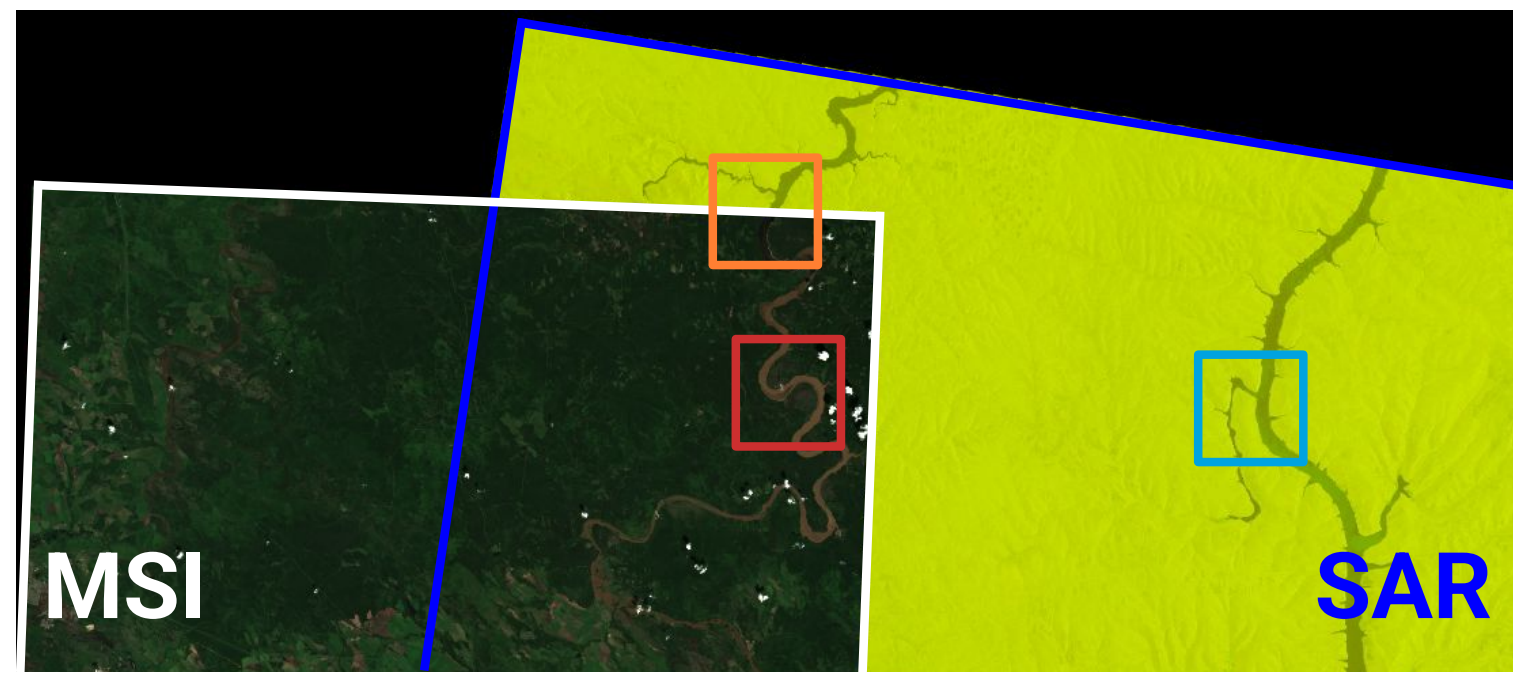


Figure 1. Example for coregistration of SAR and MSI data.

II. Data and Methods

- Cloud to Street - Microsoft Floods Dataset:**
 - Contains 900 pairs of Sentinel-1 (SAR) and Sentinel-2 (MSI) data capturing 18 global flood events.
- Proposed Model: SMAGNet (Spatially Masked Adaptive Gated Network)**

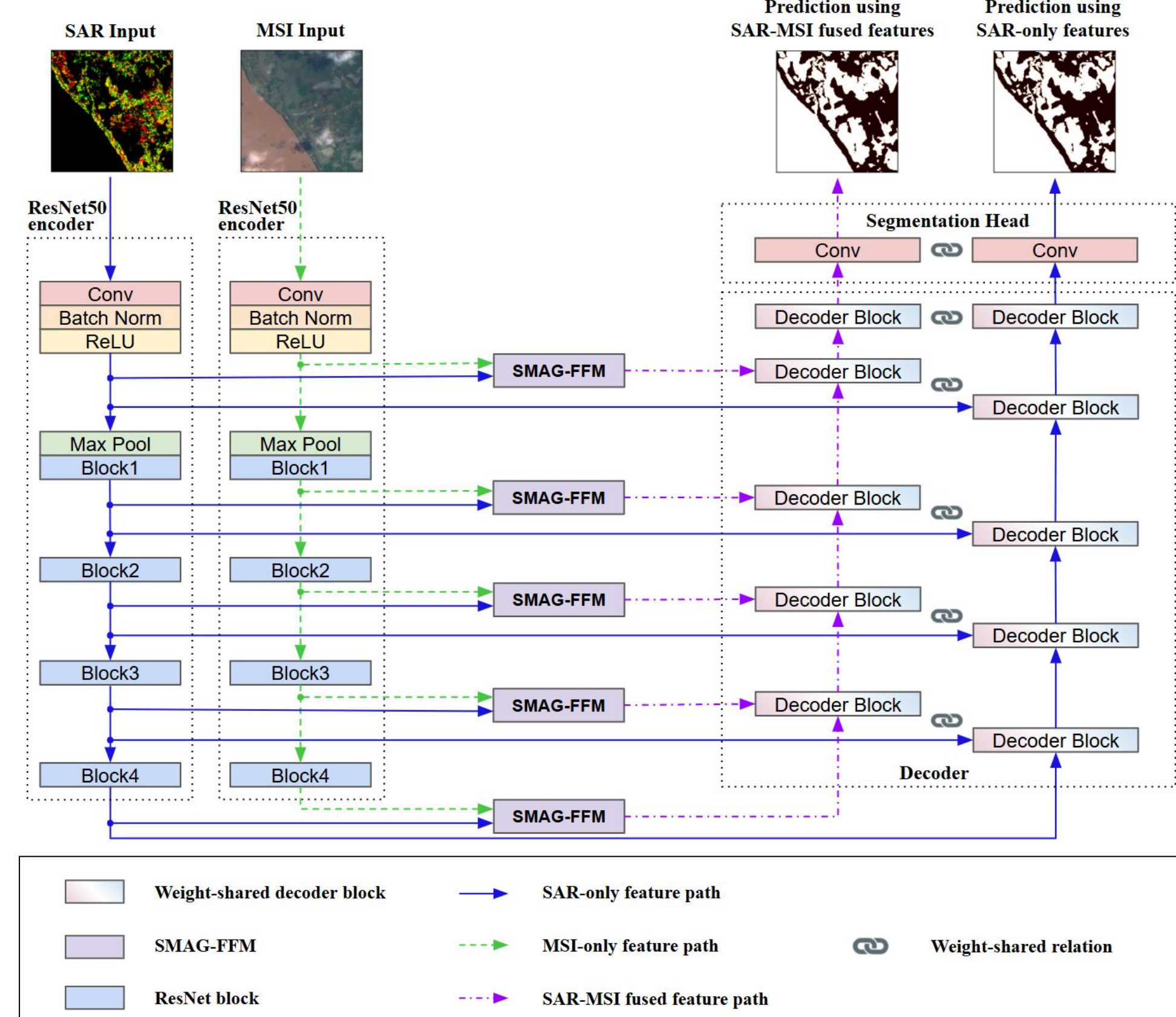


Figure 2. The architecture of SMAGNet.

III. Results

(1. Comparative Study) SMAGNet achieved the highest IoU, Recall and OA compared with other DL models.

- The comparative study evaluates the performance of SMAGNet against other multimodal deep learning models. Each experiment was conducted 10 times, and the mean and standard deviation of each metric were reported.
- In terms of IoU, SMAGNet achieved the highest performance, followed by MFGFUnet (85.96%) and MCANet (85.48%), both of which are intended to utilize SAR and MSI data as input.
- Following in IoU scores were U-Net (84.96%), CMFNet (84.95%), and CMGFNet (84.70%). CMFNet and CMGFNet are multimodal deep learning architectures specifically designed to leverage optical satellite imagery and DSM data.
- SMAGNet showed comparable performance variability to other multimodal deep learning models across four evaluation metrics.

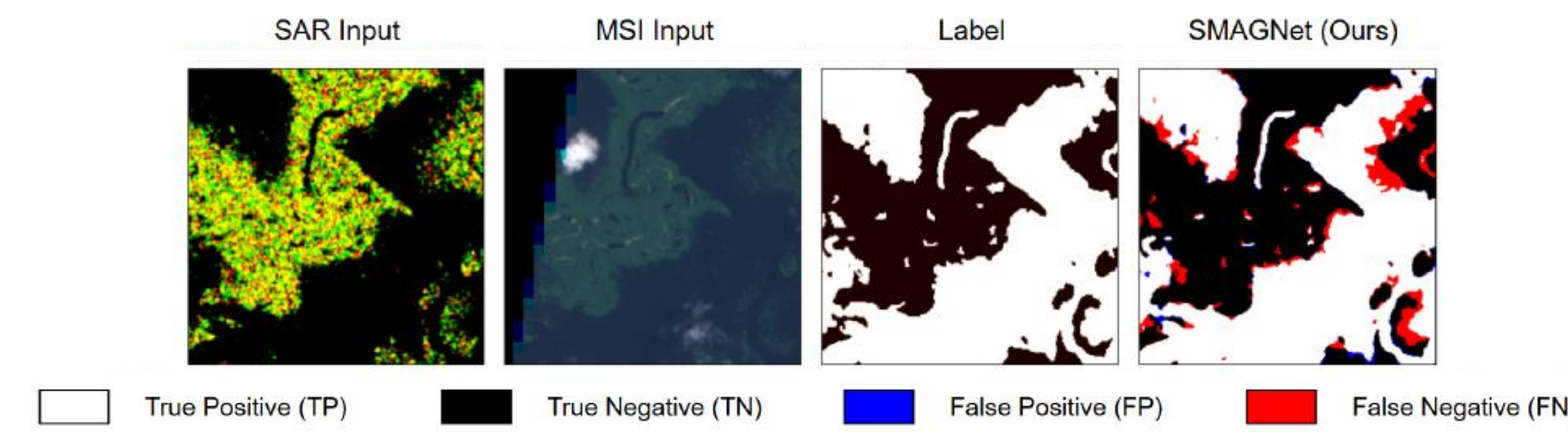


Figure 3. Visualizations of sample prediction results.

Model	IoU (%)	Precision (%)	Recall (%)	OA (%)
U-Net (SAR)	79.65 (± 0.96)	90.81 (± 0.83)	86.64 (± 1.03)	96.52 (± 0.18)
PSPNet	82.65 (± 0.85)	90.83 (± 0.93)	90.19 (± 1.29)	97.02 (± 0.15)
VFuseNet	83.33 (± 1.00)	92.98 (± 0.62)	88.92 (± 0.89)	97.20 (± 0.18)
FuseNet	83.40 (± 1.13)	92.95 (± 0.71)	89.03 (± 0.87)	97.21 (± 0.20)
FTransUNet	83.93 (± 2.64)	92.19 (± 1.08)	90.34 (± 2.46)	97.28 (± 0.47)
FPN	84.25 (± 0.96)	91.10 (± 1.03)	91.80 (± 0.54)	97.30 (± 0.19)
U-Net++	84.41 (± 1.54)	92.75 (± 0.69)	90.36 (± 1.32)	97.37 (± 0.27)
DeepLabV3+	84.48 (± 1.19)	92.04 (± 0.68)	91.14 (± 1.26)	97.37 (± 0.21)
CMGFNet	84.70 (± 0.59)	94.85 (± 0.46)	88.78 (± 0.71)	97.48 (± 0.10)
CMFNet	84.95 (± 0.87)	92.31 (± 0.93)	91.43 (± 0.95)	97.45 (± 0.16)
U-Net	84.96 (± 0.97)	92.88 (± 0.60)	90.88 (± 0.92)	97.47 (± 0.17)
MCANet	85.48 (± 0.99)	92.47 (± 0.78)	91.87 (± 0.82)	97.54 (± 0.18)
MFGFUnet	85.96 (± 0.57)	92.84 (± 0.98)	92.07 (± 0.73)	97.63 (± 0.11)
SMAGNet (Ours)	86.47 (± 0.61)	93.05 (± 0.76)	92.45 (± 0.83)	97.73 (± 0.11)

Table 1. Experimental results of the comparative study.

(2. Robustness Study) SMAGNet exhibited the highest level of robustness, consistently achieving the top IoU scores at all levels of missing data.

- The robustness study evaluates SMAGNet's ability to handle missing pixels in MSI data.
- MSI data in the test set were replaced with missing pixels at 25%, 50%, 75%, and 100%, as shown in Figure 4.
- SMAGNet maintained strong robustness, with IoU decreasing only from 86.47% with original MSI data to 79.53% at 100% missing MSI data, a minimal degradation of 6.94%.
- When MSI data is completely missing, SMAGNet maintains performance comparable to U-Net (SAR), whereas other multimodal models show significant performance degradation.

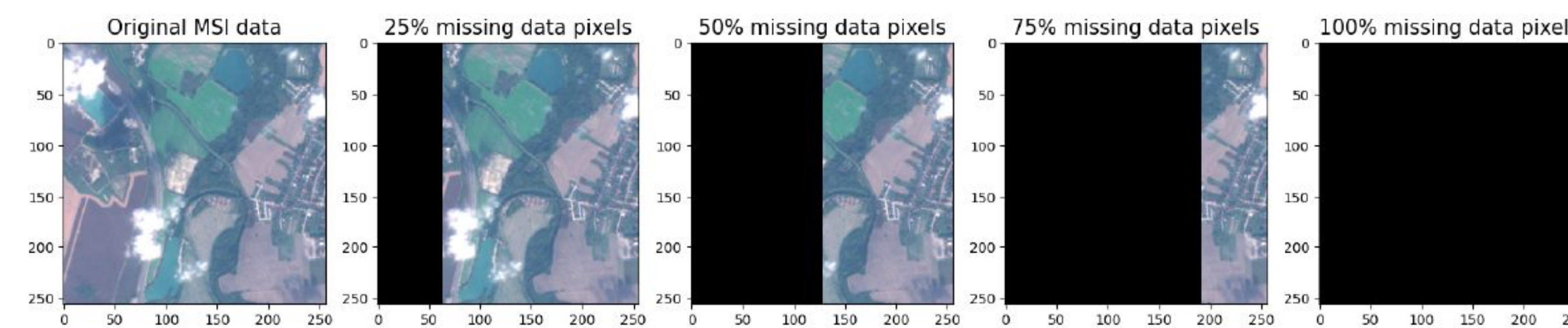


Figure 4. Visualization of sample MSI data that are replaced with missing data pixels.

Model	IoU with varying missing data pixel replacement ratios in MSI data (%)					Δ (0% - 100%)	P-value (100% missing data in MSI data vs. U-Net (SAR))
	0%	25%	50%	75%	100%		
VFuseNet	83.33 (± 1.00)	79.02 (± 1.57)	74.60 (± 2.57)	70.22 (± 3.44)	65.97 (± 3.98)	17.36	0.000 ***
FuseNet	83.40 (± 1.13)	79.49 (± 1.98)	75.36 (± 2.82)	71.23 (± 3.58)	67.24 (± 4.10)	16.16	0.000 ***
DeepLabV3+	84.48 (± 1.19)	82.43 (± 1.81)	79.22 (± 2.69)	74.19 (± 3.99)	67.33 (± 5.75)	17.15	0.000 ***
FTransUNet	83.93 (± 2.64)	78.10 (± 5.47)	75.01 (± 8.42)	72.26 (± 10.87)	67.50 (± 11.70)	16.43	0.000 ***
U-Net++	84.41 (± 1.54)	80.65 (± 2.75)	76.44 (± 4.08)	71.99 (± 5.67)	67.77 (± 6.86)	16.63	0.000 ***
PSPNet	82.65 (± 0.85)	81.16 (± 2.62)	78.10 (± 5.55)	73.60 (± 8.32)	68.40 (± 9.60)	14.25	0.000 ***
U-Net	84.96 (± 0.97)	81.62 (± 1.56)	77.83 (± 2.09)	73.89 (± 2.66)	69.97 (± 3.11)	14.99	0.000 ***
FPN	84.25 (± 0.96)	82.74 (± 1.61)	80.16 (± 2.02)	76.64 (± 2.76)	70.54 (± 6.08)	13.71	0.000 ***
MFGFUnet	85.96 (± 0.57)	83.88 (± 0.84)	81.11 (± 1.28)	77.80 (± 1.89)	72.98 (± 3.05)	12.97	0.000 ***
MCANet	85.48 (± 0.99)	83.87 (± 1.05)	81.86 (± 1.41)	78.70 (± 2.16)	74.71 (± 3.25)	10.77	0.001 ***
CMGFNet	84.70 (± 0.59)	82.37 (± 0.89)	80.32 (± 1.19)	78.17 (± 1.52)	76.34 (± 1.80)	8.36	0.000 ***
CMFNet	84.95 (± 0.87)	84.20 (± 0.65)	82.64 (± 1.12)	80.43 (± 1.50)	77.92 (± 1.21)	7.03	0.002 **
SMAGNet (Ours)	86.47 (± 0.61)	84.70 (± 0.80)	83.07 (± 0.99)	81.17 (± 1.16)	79.53 (± 1.28)	6.94	0.850

Table 2. Experimental result of the robustness study.

(3. Ablation Study) The result shows that the spatial mask and weight-shared decoder improve performance and robustness in multimodal post-flood water mapping.

- The ablation study evaluates the contributions of the spatial mask in SMAG-FFM and the weight-shared decoder to SMAGNet's performance.

Case	Weight-shared Decoder	Spatial Mask	IoU with varying missing data pixel replacement ratios in MSI data (%)					Δ (0% - 100%)
			0%	25%	50%	75%	100%	
(a)			85.77 (± 0.68)	83.29 (± 0.84)	80.85 (± 1.28)	78.10 (± 1.85)	75.86 (± 2.27)	9.91
(b)	✓		85.61 (± 0.86)	83.58 (± 0.86)	81.52 (± 0.95)	79.30 (± 1.22)	77.11 (± 1.42)	8.50
(c)	✓	✓	86.47 (± 0.61)	84.70 (± 0.80)	83.07 (± 0.99)	81.17 (± 1.16)	79.53 (± 1.28)	6.94

Table 3. Experimental result of the ablation study.

IV. Conclusion

- We introduce a novel Spatially Masked Adaptive Gated Network (SMAGNet) to adaptively integrate partially available MSI data into the SAR based post-flood water mapping process based on multimodal deep learning.
- We demonstrate the superior performance of SMAGNet in post-flood water mapping compared to other multimodal deep learning models through comprehensive experimental results.
- Furthermore, we found that even when MSI data were completely missing, the performance of SMAGNet remained statistically comparable to that of a U-Net model trained solely on SAR data.
- This finding indicates that our method enhances the model robustness to missing value pixels in MSI data and applicability of multimodal deep learning in real-world flood management scenarios.

Reference

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- Konapala, G., Kumar, S. V., & Ahmad, S. K. (2021). Exploring Sentinel-1 and Sentinel-2 diversity for flood inundation mapping using deep learning. *ISPRS Journal of Photogrammetry and Remote Sensing*, 180, 163-173.

Supplementary Materials

- SMAG-FFM (Spatially Masked Adaptive Gated Feature Fusion Module)**
 - Spatially Masked Gating: Selectively accepts valid MSI features using a spatial mask.

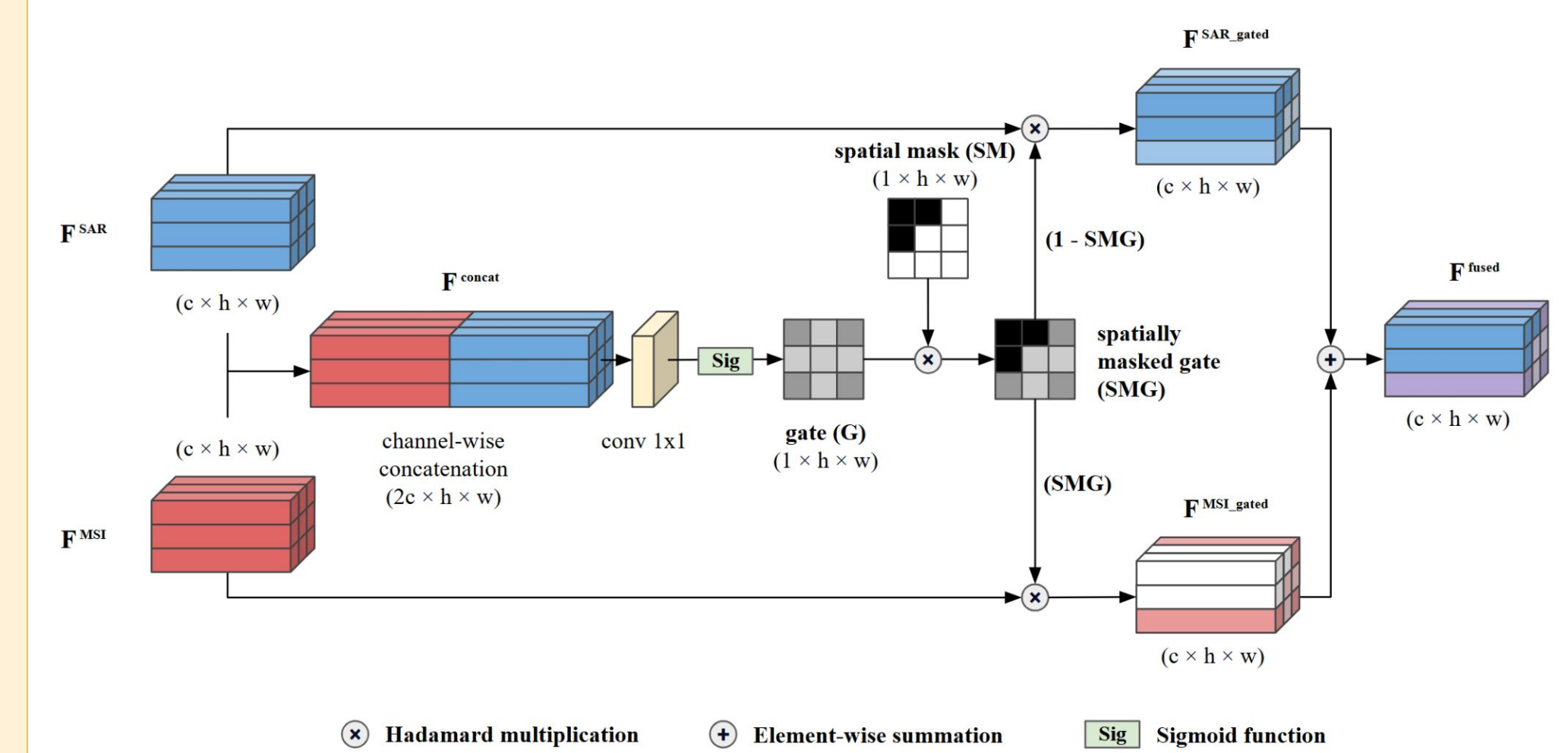


Figure 5. Structure of the SMAG-FFM.